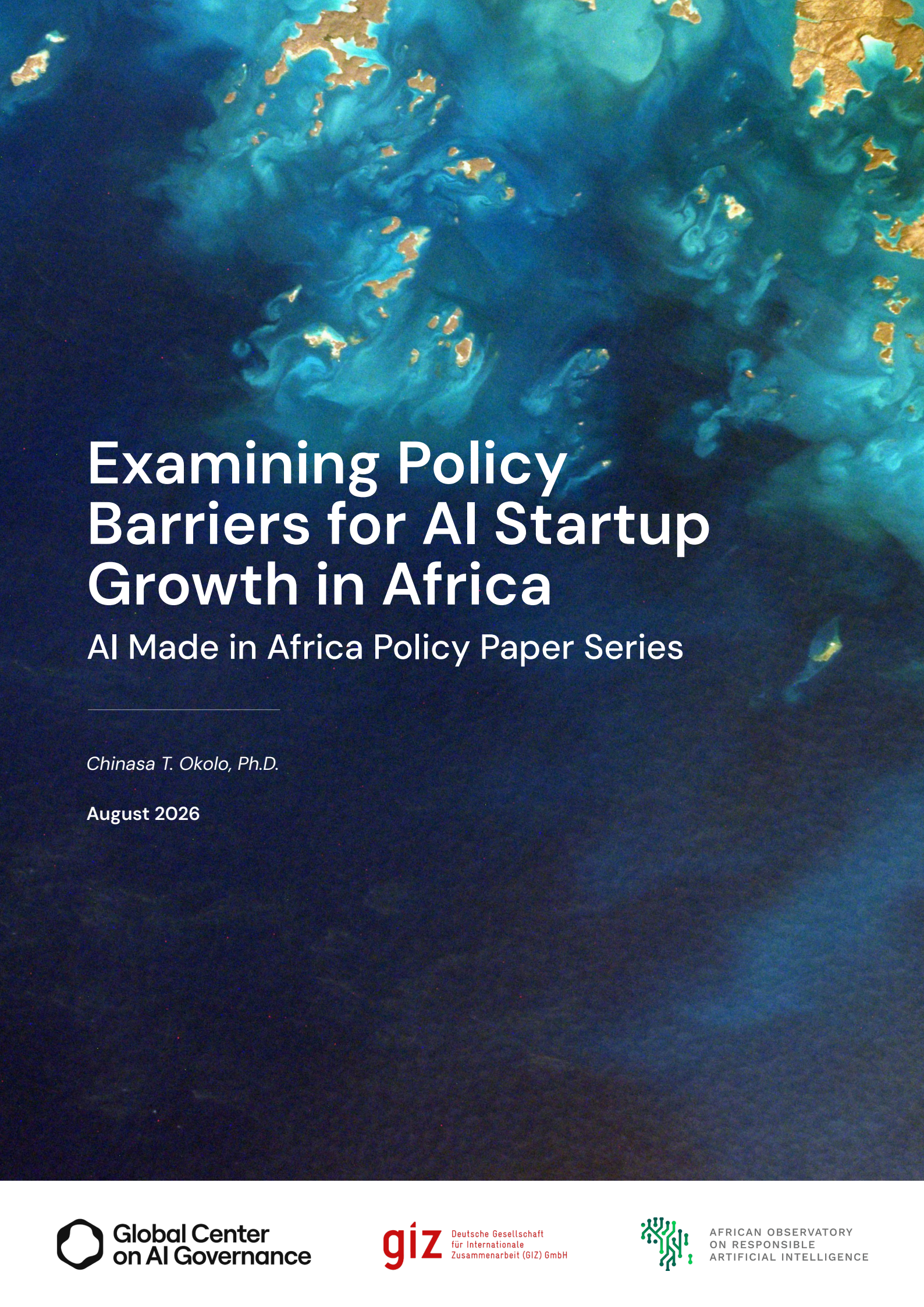




Examining Policy Barriers for AI Startup Growth in Africa

AI Made in Africa Policy Paper Series

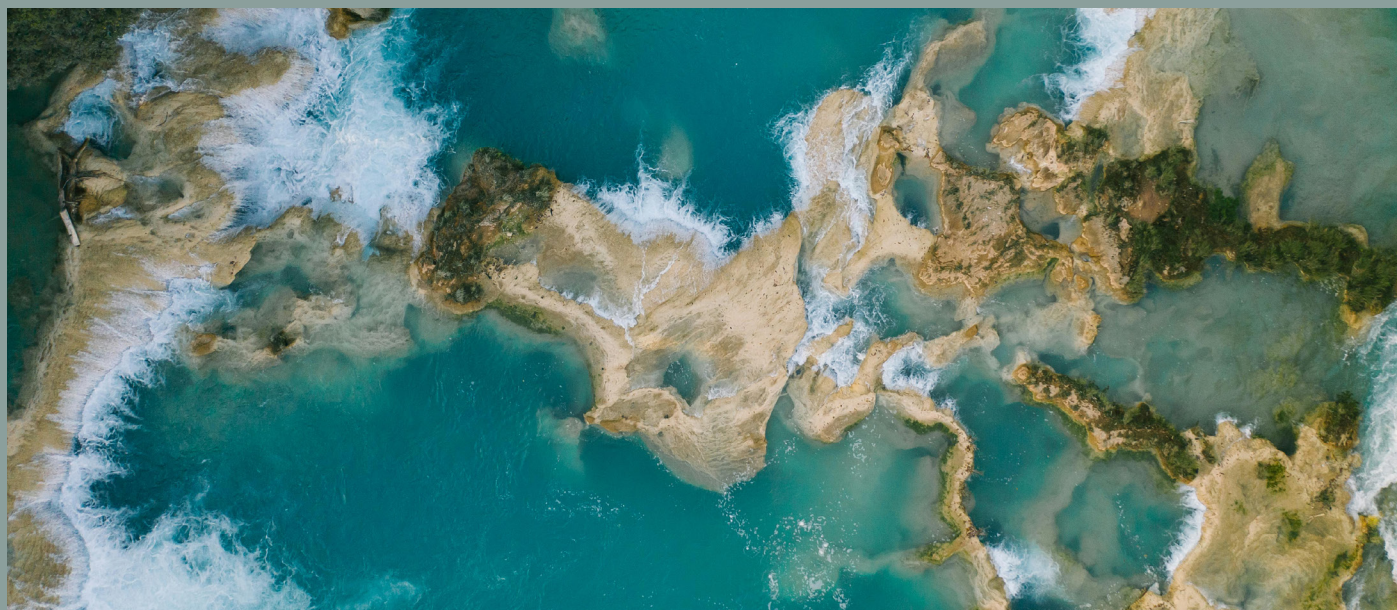
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Author Biography and Acknowledgements

Chinasa T. Okolo, Ph.D., is the Founder of Technecultura and a recent Computer Science Ph.D. graduate from Cornell University. Her research focuses on AI governance and safety for the Global Majority, datafication and algorithmic marginalization, and the geopolitical impacts of AI. Dr. Okolo has been recognized by TIME as one of the world's most influential people in AI, honored on the inaugural Forbes 30 Under 30 AI list, and advises numerous multilateral institutions, national governments, corporations, and nonprofits. She has previously worked as a World Bank Group Africa Fellow in the Development Economics Research Group, a Policy Specialist at the United Nations Office for Digital and Emerging Technologies (ODET), a Fellow at the Brookings Institution, and in research-based roles at Apple and Microsoft. Her research has been widely covered in media outlets and published at top-tier venues in human-computer interaction and sociotechnical computing.

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Executive Summary

Despite considerable policy attention devoted to AI development across Africa, the ecosystem conditions necessary for startup-driven innovation are largely overlooked, underexamined, and undersupported. Small and medium-sized enterprises (SMEs) that build and adopt AI solutions face a compounding set of structural barriers that existing frameworks have largely failed to address. These include regulatory ambiguity, fragmented digital economy legislation, shallow capital markets, infrastructure deficits, and the near-absence of continent-specific AI technical standards. This paper draws on the Africa AI Startups Database, a resource cataloging 224 AI startups operating across nearly two dozen African countries, to examine how national AI strategies, startup-focused digital economy frameworks, and emerging standardization initiatives either enable or constrain startup-led AI innovation. This analysis posits that most AI and digital economy frameworks remain aspirational rather than operationally enabling for early-stage ventures. It provides key policy recommendations focused on codifying startup acts, revising national AI strategies, accelerating AI standards development in Africa, establishing shared AI infrastructure, expanding AI-specific startup incubation programs, and encouraging academic spinouts.



1. Introduction

Africa's artificial intelligence (AI) startup ecosystem is both dynamic and fragile. Across at least 22 countries, small and medium-sized enterprises are deploying AI technologies to address fundamental development challenges in healthcare, agriculture, financial services, and education. Yet African policy frameworks have yet to systematically address the structural barriers that constrain small-scale innovators, who are the very actors positioned to deliver contextually relevant AI solutions. These ventures face precarious operating conditions, forcing them to navigate chronic infrastructure deficits that span unreliable digital connectivity, fragmented payment systems, shallow capital markets, acute technical talent shortages, and regulatory ambiguity. High-profile startup failures underscore how these compounding barriers create existential risks that policy interventions and tailored standards could meaningfully address.

Understanding and dismantling these barriers matters not simply for increasing the competitiveness of individual firms, but for the broader ambition of ensuring that AI development on the continent reflects African contexts, priorities, and values. Entrepreneurial ventures have substantially catalyzed AI advancement in technologically more advanced nations, and the opportunity to replicate this dynamic on the African continent depends on whether the conditions enabling startup growth can be deliberately constructed. If early-stage ventures cannot survive long enough to mature, the innovation gap will be filled by technology designed and governed elsewhere, foreclosing the opportunity for locally grounded AI solutions. This paper examines 20 national AI strategies, regional and national digital

economy frameworks, including 10 startup-focused acts and bills, and several emerging technical standardization initiatives to identify the specific policy and regulatory barriers impeding startup-led AI innovation across Africa. It provides an evidence-based framework for governments seeking to translate strategic ambitions into startup-enabling conditions, explores opportunities for academic-industry linkages, and offers actionable recommendations to strengthen existing national AI strategies and inform forthcoming AI legislation. By grounding policy discourse in empirical patterns of African AI entrepreneurship, this work contributes to regional capacity-building and to broader debates about innovation policy in resource-constrained contexts.



2. The Policy Challenge

2.1 The African AI Startup Landscape

An analysis of 224 AI startups operating across at least 22 African countries reveals an ecosystem defined by geographic concentration, sectoral depth, and a pronounced scarcity of mature firms. Nigeria (50 companies), South Africa (49), and Kenya (34) collectively account for over 130 startups, representing more than 60 percent of the surveyed landscape. Ghana, Tunisia, and Egypt contribute a further 42 companies, while emerging ecosystems in Rwanda, Uganda, Morocco, and Ethiopia signal growing geographic diversification (Okolo, 2026a).

Sectorally, the landscape reflects a markedly different orientation from that of AI industries in more economically developed markets. While software development is the largest domain, with 74 companies producing software tools for customers or other businesses, there is substantial activity in healthcare (24), finance (23), and agriculture (23). Rather than prioritizing consumer entertainment or advertising applications, African AI entrepreneurs are deploying technology to address fundamental development challenges in food security, healthcare delivery, and financial inclusion. This sectoral orientation offers policymakers a clear roadmap for targeted interventions and indicates that AI in Africa functions largely as a development tool rather than a consumer product.

The ecosystem's most consequential structural vulnerability is a pronounced bottleneck at the scaling phase. Of the 224 documented startups, 143 remain early-

stage ventures, 72 have reached growth stage, and only 9 have attained maturity. This reflects chronic deficits in late-stage venture capital, cross-border regulatory fragmentation that constrains market expansion, and a continent-wide shortage of AI technical talent (Okolo, 2026b). Africa accounts for approximately 1 percent of the world's AI talent, with total professional capacity estimated between 5,000 and 62,000 individuals (JICA, 2025; Cisco & Carnegie Mellon University Africa, 2025). The capital gap is equally stark: in the third quarter of 2025, African AI startups received just 0.03 percent of global venture capital investment in AI, amounting to \$14 million of the \$47.8 billion raised globally, compared with 81.2 percent flowing to United States companies during the same period (Kanife, 2025).

2.2 Startup and Digital Economy Bills

Several African governments have enacted or introduced legislation aimed at formalizing entrepreneurial ecosystems. These include the Senegalese Startup Act, the Tunisian Startup Act, the Nigerian Startup Act, the Ethiopia Startup Proclamation, the Mauritania Startup Act, and proposed frameworks in Togo, Kenya, Ghana, and Mali (Rodrigues, 2021; Dosso, 2024). At the continental level, the African Union has developed a Startup Policy Framework and Model Law to provide harmonized guidance for member states on accelerating innovation ecosystem growth (African Union, 2024).

These instruments introduce meaningful mechanisms, including simplified startup registration, tax incentives, and in some cases pathways to public procurement. Yet their translation into operational reality for AI startups remains limited. Few explicitly address the technical and financial prerequisites specific to AI development, such as data governance, compute infrastructure access, or algorithmic accountability standards. Implementation capacity and dedicated funding remain inconsistent across jurisdictions, creating a gap between legislative intent and on-the-ground practice.

The risks of poorly calibrated regulation are visible in documented startup disruptions. In Nigeria, a municipal policy restricting motorcycle-hailing directly contributed to companies like Gokada laying off 80 percent of its workforce, Max terminating approximately 2,000 riders, and ORide suspending operations entirely (Adeoye & Idris, 2021). A separate directive from the Central Bank of Nigeria, issued in February 2021 and later lifted in December 2023, restricted banks from servicing cryptocurrency exchanges, cut fintech startups off from the formal financial system, and froze customer accounts (Osazuwa et al., 2024). Instead of curtailing crypto transactions, the measure pushed trading onto unregulated peer-to-peer platforms, reducing government oversight and tax revenue while increasing consumer risk (Mehta, 2025). These cases illustrate how blunt or inadequately calibrated regulatory instruments can destabilize active startup sectors rather than support them, a risk that grows as AI systems become more deeply embedded in critical services.

2.3 National AI Strategies

At least 20 African nations have adopted national AI strategies, including Algeria, Angola, Benin, Cameroon, Côte d'Ivoire, Egypt,

Ethiopia, Ghana, Kenya, Lesotho, Mauritania, Mauritius, Namibia, Nigeria, Rwanda, Senegal, South Africa, Tunisia, Zambia, and Zimbabwe (Okolo & Raji, 2026). The AU Continental AI Strategy provides an overarching framework for member states to unify national regulatory approaches and advance the socioeconomic benefits of AI (African Union, 2024b). Together, these documents represent substantial institutional investment in AI governance and signal a growing recognition of AI as a development priority.

Yet most of these strategies address AI at the sectoral and macroeconomic levels without descending into the operational conditions that determine whether early-stage ventures can survive and grow. References to SMEs and startups tend to appear as rhetorical commitments rather than as the basis for specific, funded policy instruments. Questions of startup financing mechanisms, regulatory sandboxes, AI procurement preferences, and talent pipeline development are rarely addressed in systematic detail. This results in a structural gap between strategic intent and the enabling environment that AI entrepreneurs across the continent encounter in reality. Addressing this gap requires a deliberate reorientation of existing frameworks toward the conditions that enable startups to move from inception to maturity.

2.4 AI Standardization

Alongside regulatory frameworks, technical standardization represents an underutilized lever for enabling responsible AI expansion in Africa. While international bodies, including the International Organization for Standardization (ISO) and the IEEE Standards Association (IEEE-SA), have advanced global AI standardization efforts, the African Organisation for Standardisation (ARSO) has yet to independently finalize standalone technical AI standards adapted to the continent's diverse regulatory and market contexts.

A significant recent development is the ISO-ARSO Kigali Agreement, signed in October 2025, which establishes a framework to strengthen technical cooperation and accelerate the adoption and impact of international standards across Africa (ISO, 2025). This work is preceded by the African Fourth Industrial Revolution Standardization Strategy, developed in partnership with the IEEE-SA, which aims to align regional technology standards across areas such as AI and blockchain to foster a unified African digital economy under the AfCFTA framework (IEEE-SA & ARSO, 2021). This agreement and strategic roadmap provide a foundation for ARSO to translate global AI standards into African contexts. Other efforts, like the Africa Tech Index (AT50)¹, launched at the London Stock Exchange in January 2026, introduce rules-based benchmarking and governance transparency to Africa's private technology sector, and can be used to reinforce momentum toward market and technical discipline for African AI startups (IndexaExchange, 2026).

The 2025 World Development Report on Standards for Development, which advocates that standardization is a foundational enabler of economic growth, consumer protection, and market integration (World Bank Group, 2025), can also be leveraged to advance standardization as regulatory AI frameworks continue to expand across Africa. For AI startups specifically, standards reduce market entry barriers by clarifying compliance expectations, building trust with institutional partners and development finance institutions, and creating more predictable operating environments. Integrating AI standards into national strategies and startup-enabling legislation would strengthen the overall

policy architecture, offering a complement to context-sensitive regulation that promotes both market discipline and innovation.



¹ Indexa Exchange Group, 'African Tech Index | AT50 Index' (Indexa Exchange Group, 2025) <https://www.indexaexchange.com/> accessed 23 May 2026.

3. Policy Recommendations

The evidence presented above reveals a coherent structural challenge: Africa's AI policy architecture is building ambition without the operational foundation to translate it into startup-enabling conditions. The following recommendations are directed at national governments, the African Union, regional economic communities (RECs), and development partners.

Scale AI-Specific Incubation Programs and Entrepreneurship Support

Several existing programs illustrate the potential of targeted support for AI startups. Nigeria's iHatch program², Egypt's StartIT Incubation Program³, Cabo Verde's Tech4Good Hub Incubation Program⁴, Rwanda's Innovate Rwanda platform⁵, the UNDP AI Hub for Sustainable Development⁶, and Google for Startups Africa Accelerator⁷ collectively demonstrate the value of structured programming. Governments, development partners, and industry should commit to establishing dedicated AI-focused incubation tracks that provide mentorship, compute access, and pathways to government procurement, addressing the full lifecycle of a startup rather than focusing solely on early-stage formation.

Codify Startup Acts into Enforceable, Funded Legislation with AI-Specific Provisions

Startup acts across the continent must move beyond symbolic startup recognition toward substantive operational support. This requires dedicated funding mechanisms, including models such as public-private partnerships or sovereign wealth fund allocations; specific provisions addressing AI data governance and intellectual property; and simplified licensing regimes for AI-enabled services. The AU Startup Policy Framework and Model Law offers a useful template that national legislators should draw on as they finalize pending startup bills in Kenya, Ghana, Togo, and Mali, ensuring that AI-specific considerations are integrated from the outset and that regulatory design is tested against the reality of startup operations before enactment.

² Office for Nigerian Digital Innovation, 'iHatch Portal' (ONDI Portal) <https://ondi.nitda.gov.ng/#/ihatch> accessed 27 May 2026.

³ Technology Innovation and Entrepreneurship Center, 'StartIT Incubation Program' (TIEC Portal) <https://tiec.gov.eg/English/Programs/Start-IT/Pages/default.aspx> accessed 27 May 2026.

⁴ Tech4Good Cabo Verde, 'Incubation Program' (Tech4Good Hub, 2026) <https://tech4good.cv/incubation-program/> accessed 27 May 2026.

⁵ Innovate Rwanda, 'Innovate Rwanda 2.0' (Innovate Rwanda Portal, 2025) <https://innovaterwanda.rw/en> accessed 27 May 2026.

⁶ AI Hub for Sustainable Development, 'Reimagining AI Partnerships' (AI Hub for Sustainable Development Portal, 2026) <https://aihubfordevelopment.org/> accessed 27 May 2026.

⁷ Google for Startups, 'Google for Startups Accelerator: Africa' (Google for Startups Portal, 2026) <https://startup.google.com/programs/accelerator/africa/> accessed 27 May 2026.

Revise National AI Strategies to Address the Startup Enabling Environment

Most national AI strategies and draft laws across Africa offer insufficient guidance on the conditions that enable early-stage AI ventures to grow. Updated strategy cycles should incorporate specific commitments to startup financing instruments, regulatory sandboxes, talent pipeline programs, and AI procurement initiatives that prioritize SME participation. This revision should be developed through direct consultation with the startup ecosystem, not solely through government and academic channels, ensuring that strategic ambitions are grounded in the operational realities that founders and early-stage teams actually face.

Encourage Academic Spinouts and Strengthen University-to-Market Talent Pipelines

The talent shortage constraining Africa's AI startup sector is partly a structural byproduct of weak linkages between academic AI research and commercial application. Governments should incentivize universities to establish technology transfer offices, introduce student founder programs, and create equity participation frameworks that allow academic institutions to share in spinout success, replicating models that have accelerated innovation in the United States, Canada, and the United Kingdom. Building these pipelines will expand both talent supply and the quality of AI solutions reaching the market. Policies should also create tax incentives that encourage African talent emerging from academic institutions to receive competitive compensation to work for local startups, rather than remote international firms, as a way to reduce the brain-drain dynamic that limits long-term capacity.

Establish Regional, Shared AI Infrastructure

The cost of compute, data storage, and testing environments constitutes a significant barrier for early-stage startups operating in markets with limited digital infrastructure. Regional cooperation mechanisms, potentially coordinated through the AU or RECs, including ECOWAS, EAC, SADC, ECCAS, and AMU, should explore shared AI infrastructure. Distributing costs across multiple national stakeholders would make participation viable for smaller economies and accelerate the geographic diversification of Africa's AI startup ecosystem beyond its current concentration in a handful of dominant markets.

Advance AI Standardization Through the ISO-ARSO Kigali Agreement

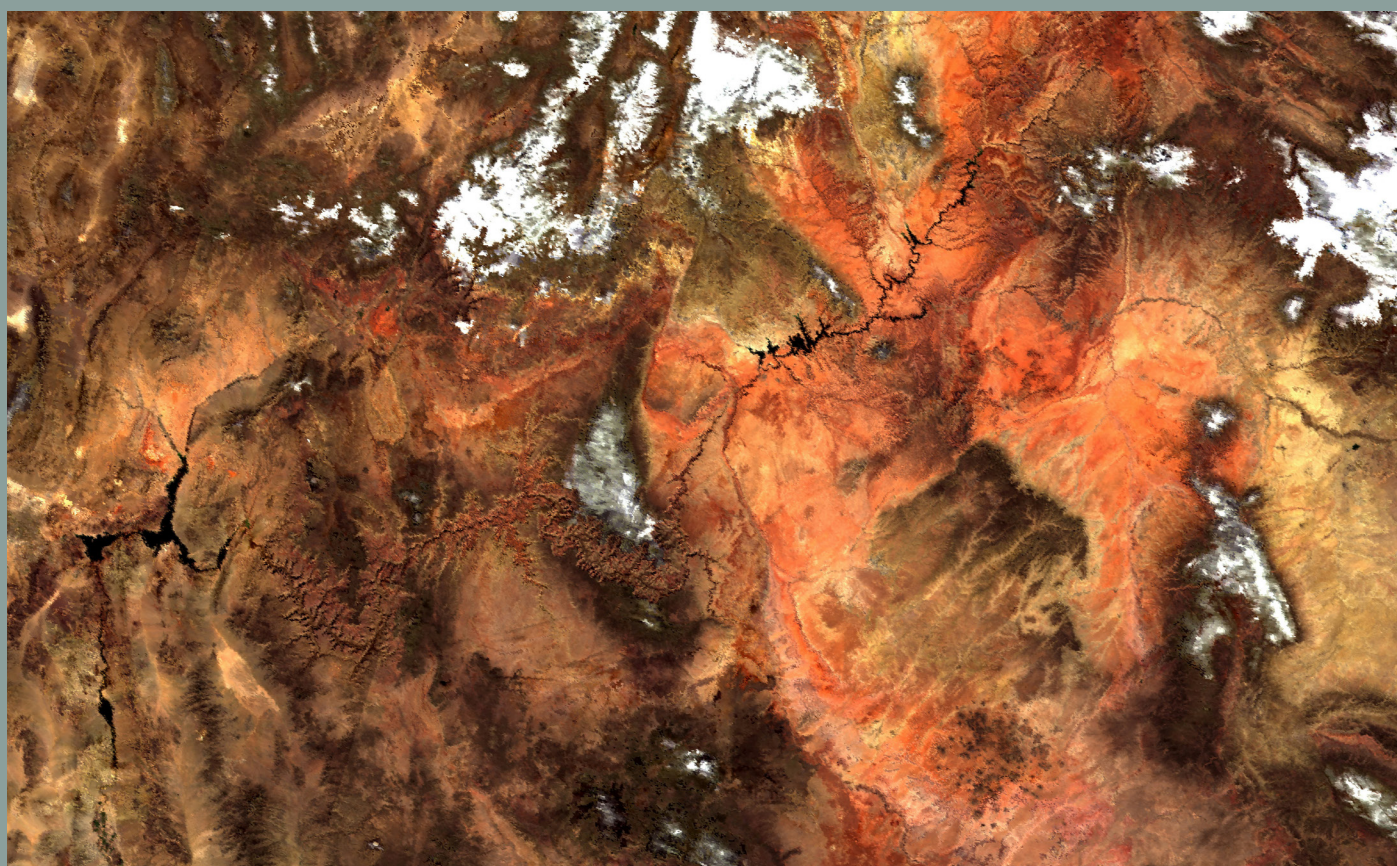
ARSO should leverage the Kigali Agreement and African Fourth Industrial Revolution Standardization Strategy to develop AI standards adapted to African market conditions, informed by the 2025 World Bank Group World Development Report on Standards for Development and the governance benchmarks introduced by the Africa Tech Index. These standards should also advocate for startups to advance multilingual AI, resource-efficient models, algorithmic transparency, and interoperability across African markets, thereby aligning AI development with the continent's distinct regulatory and development priorities. Development partners and bilateral donors should provide technical assistance to support ARSO's capacity to develop and implement these standards, creating a consistent compliance environment that reduces barriers for startups seeking to scale across borders and improves regulatory coherence across the continent.

Conclusion

Africa's AI startup ecosystem has demonstrated genuine resilience and sectoral purpose. The concentration of AI ventures in healthcare, agriculture, finance, and education reflects an entrepreneurial orientation toward solving real development challenges, a characteristic that distinguishes the continent's AI sector from its counterparts in more economically advanced markets. This is a distinctive asset that deserves deliberate policy protection and institutional support.

However, the structural conditions required to translate early-stage activity into mature, scalable enterprises remain inadequate. National AI strategies have largely failed to address the operational reality of startup growth. Digital economy legislation, while proliferating, is inconsistently implemented, insufficiently targeted to AI-specific needs, and, in some cases, has been calibrated in ways that destabilize rather than support the startup sector. AI standardization at the continental level remains nascent, but provides an ideal opportunity to strengthen AI products and services.

Addressing these conditions requires coordinated action from governments, the African Union, RECs, and development partners. The foundational architecture is beginning to take shape through ARSO standardization efforts, the AU Startup Model Framework, and emerging national startup acts, which together constitute a meaningful starting point. African stakeholders must also provide the political will and institutional capacity to ensure these frameworks translate into startup-enabling conditions on the ground. If successful, Africa's AI entrepreneurs are well-positioned to deliver contextually grounded solutions at scale.



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